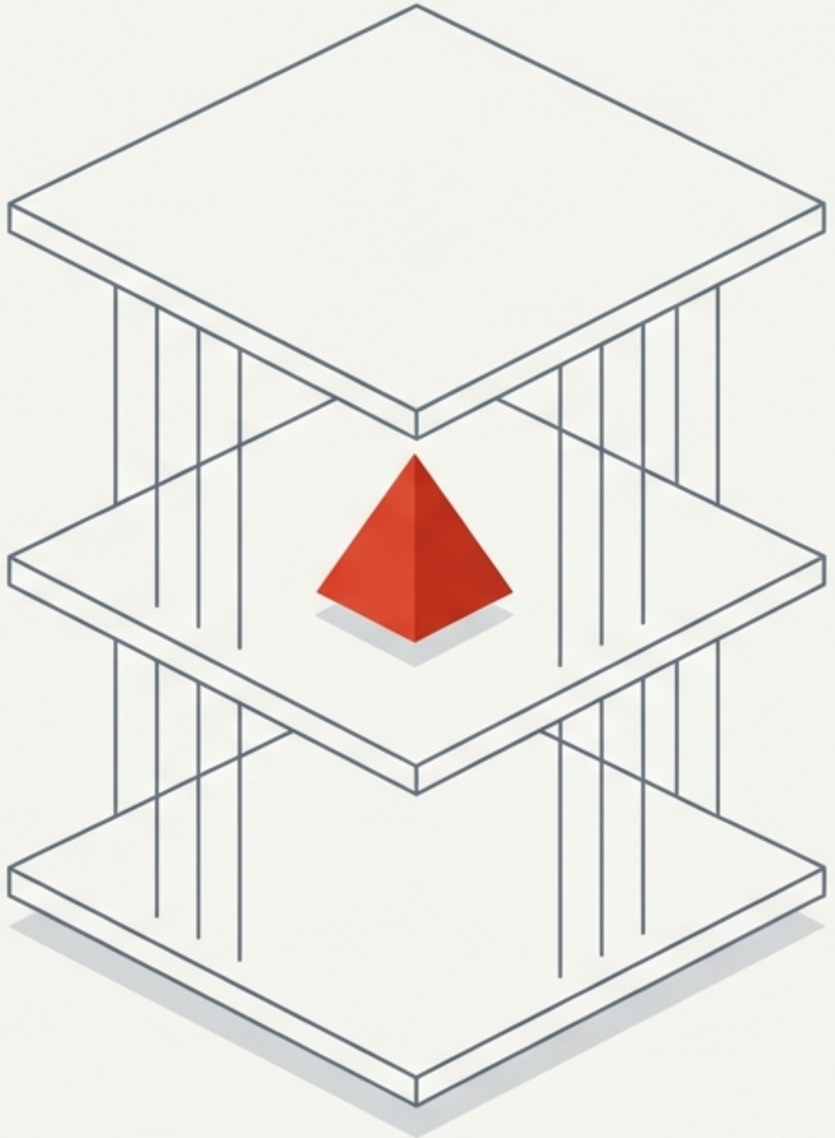
SEV-1: The Capstone Diagnostic

AWS Day 2 Operations & Troubleshooting • NorthBank • Lab 15

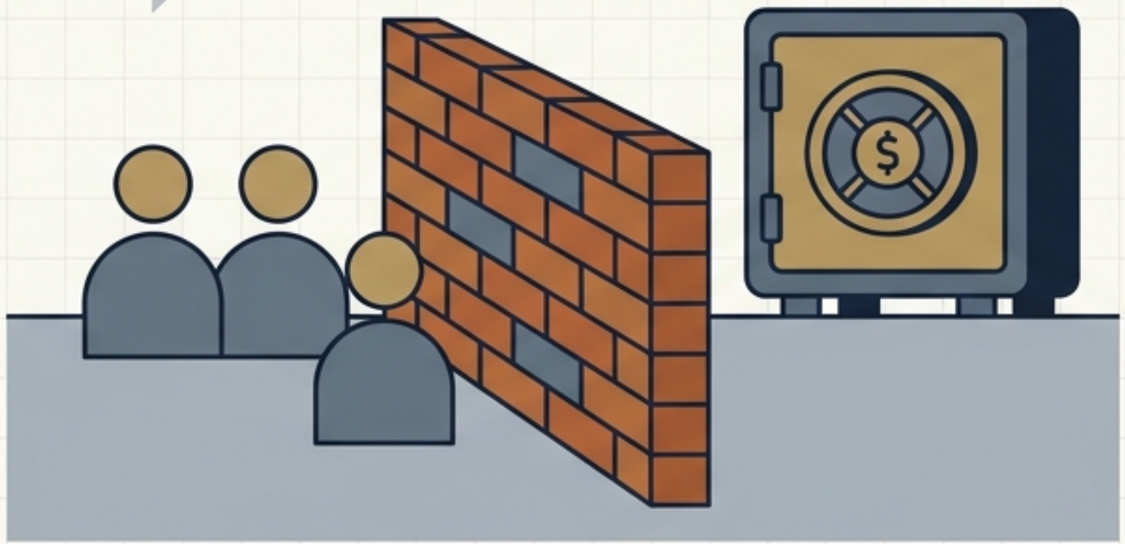
Priority: SEV-1
Impact: Total Outage
Environment: Production



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Incident Impact Card

Internet banking is DOWN. Customers can't log in or move money.



Severity: SEV-1

Service:
NorthBank Internet Banking

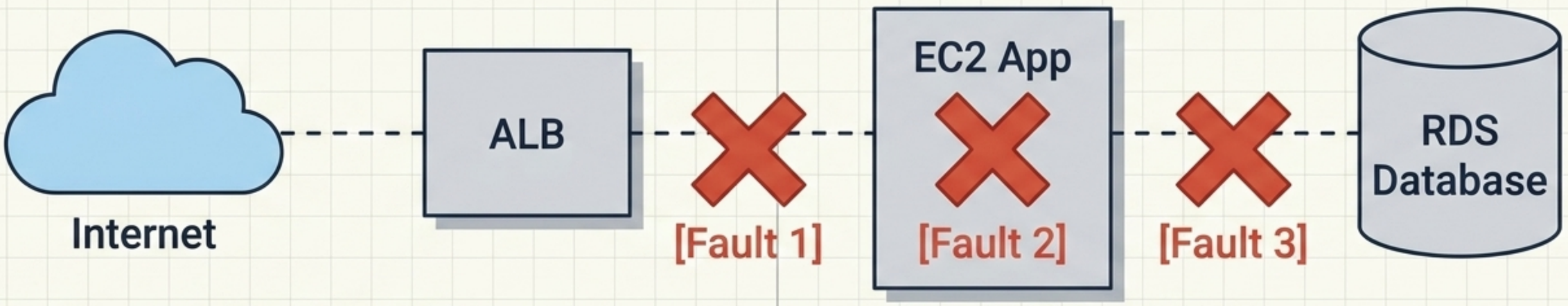
User Impact:
100% Login Failure & Transaction Block

Root Cause Profile:
Multiple / Chained Faults Detected

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Technical Operations Playbook

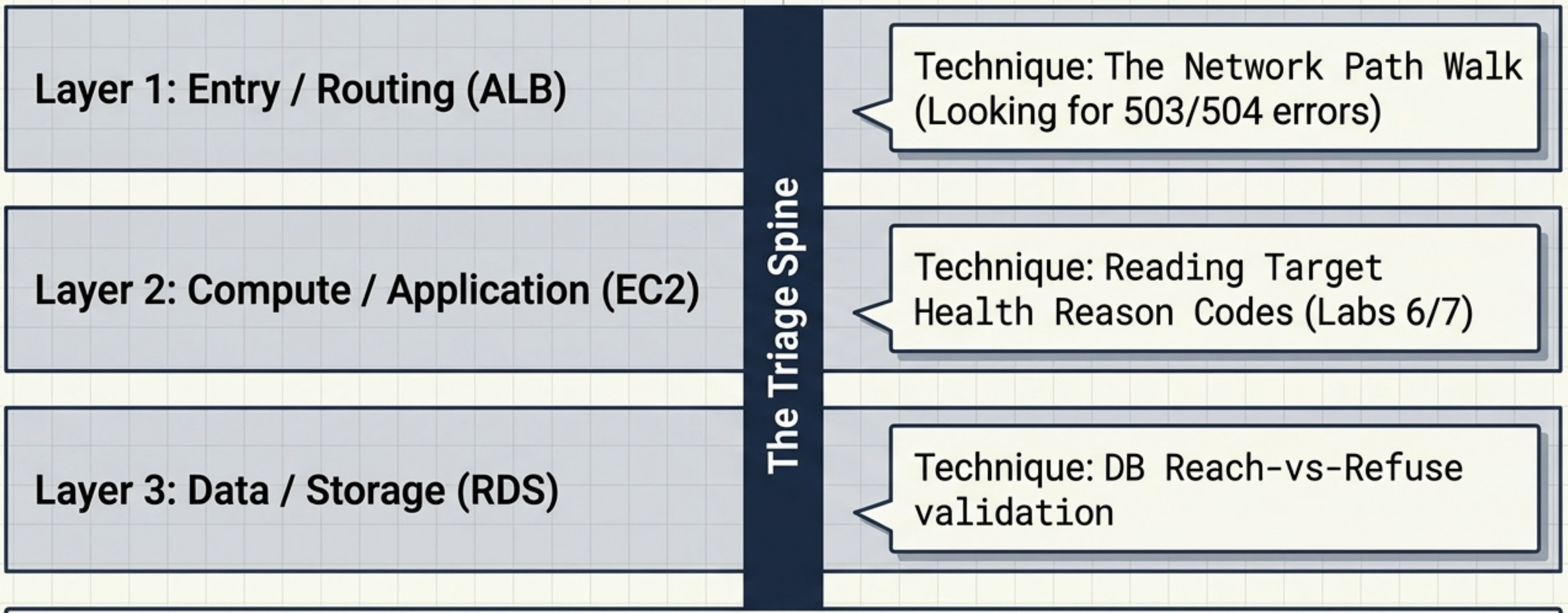
```
$ export APP_SG, ALB_SG, DB_SG, TG_ARN from terraform output
$ ./break.sh
```



Script execution initiates automated chained faults across infrastructure.

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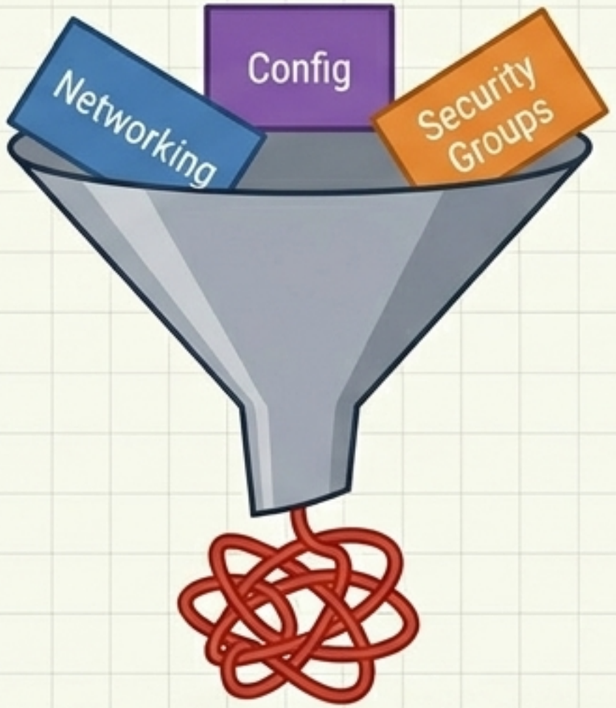
The Triage Spine



The Triage Spine scales: it is the same logical flow, just walked multiple times as we peel back the layers.

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The Chaos Trap



Unable to Determine Root Cause / Risk of Cascading Failure

One Hypothesis at a Time

Isolate one variable.

↓

Apply test/fix.

↓

Verify state change.

✓ Clear, Verified Result

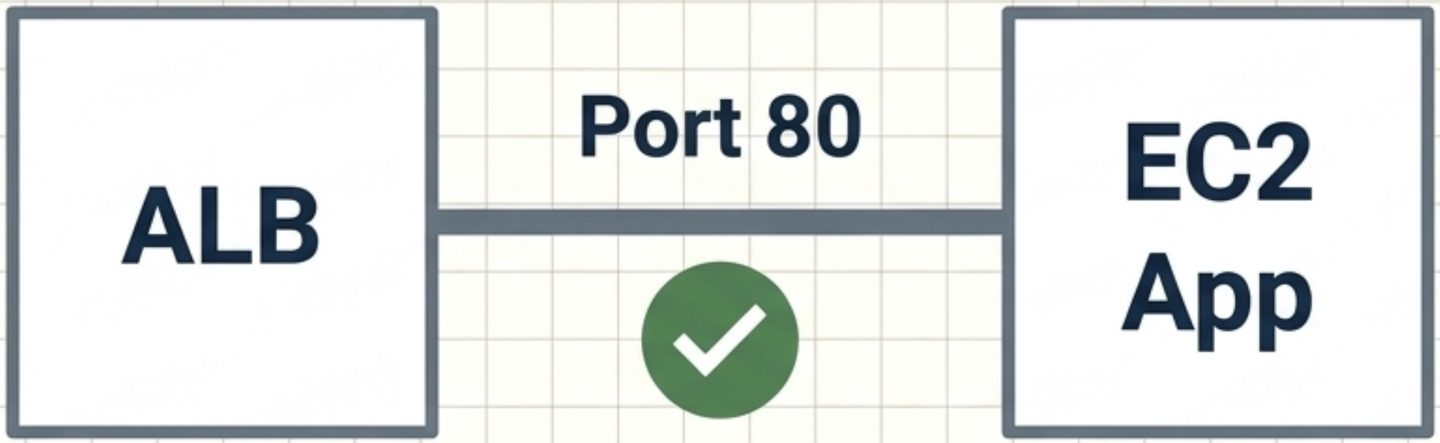
Fixing the first thing rarely fixes everything in a chained fault. If you change three things at once, you lose the ability to diagnose. Don’t make it worse: One hypothesis at a time, always.

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Chained Fault Diagnostic Matrix

Symptom	Hypothesis	Fix Applied	Resulting State
[503/504 Fully Down]	[Network Block]	[Authorize ALB→App Port 80]	[State Change: Still Unhealthy]
[Still Unhealthy]	[Bad Health Check]	[Fix Health Check Path]	[State Change: Site 200 OK, but no data]
[DB Cut Off]	[Backend Network Block]	[Authorize App→DB Port 3306]	[State Change: Fully Resolved]

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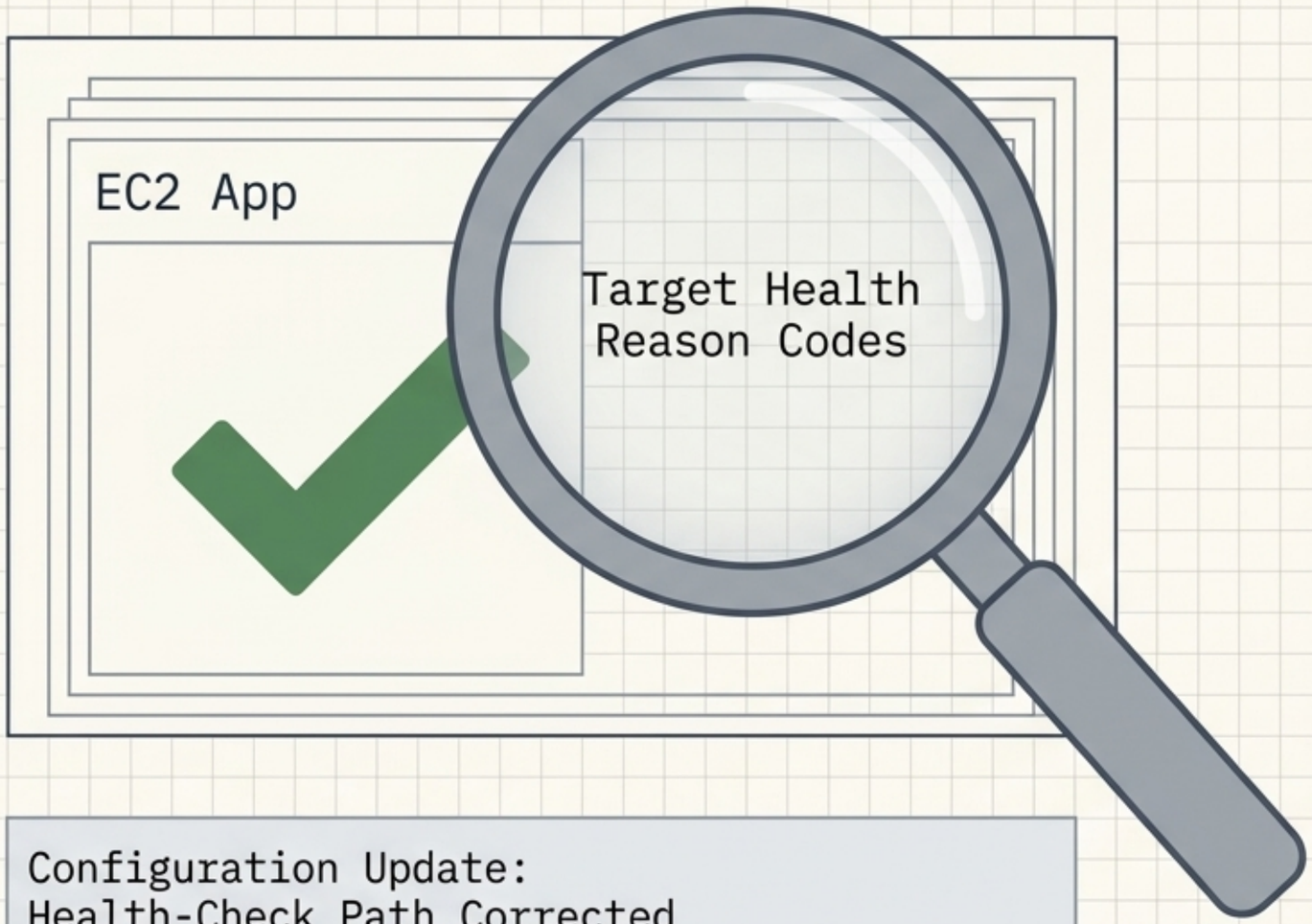

```
Authorize: ALB_SG -> APP_SG (Port 80)
```

Traffic Flowing

New State:
Still Unhealthy

Phase 1: Piercing the 503/504 Network Block.
The site was completely unreachable. Authorizing ALB to App on Port 80 restores basic connectivity, but reveals the next layer of the chained fault.

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Configuration Update:
Health-Check Path Corrected

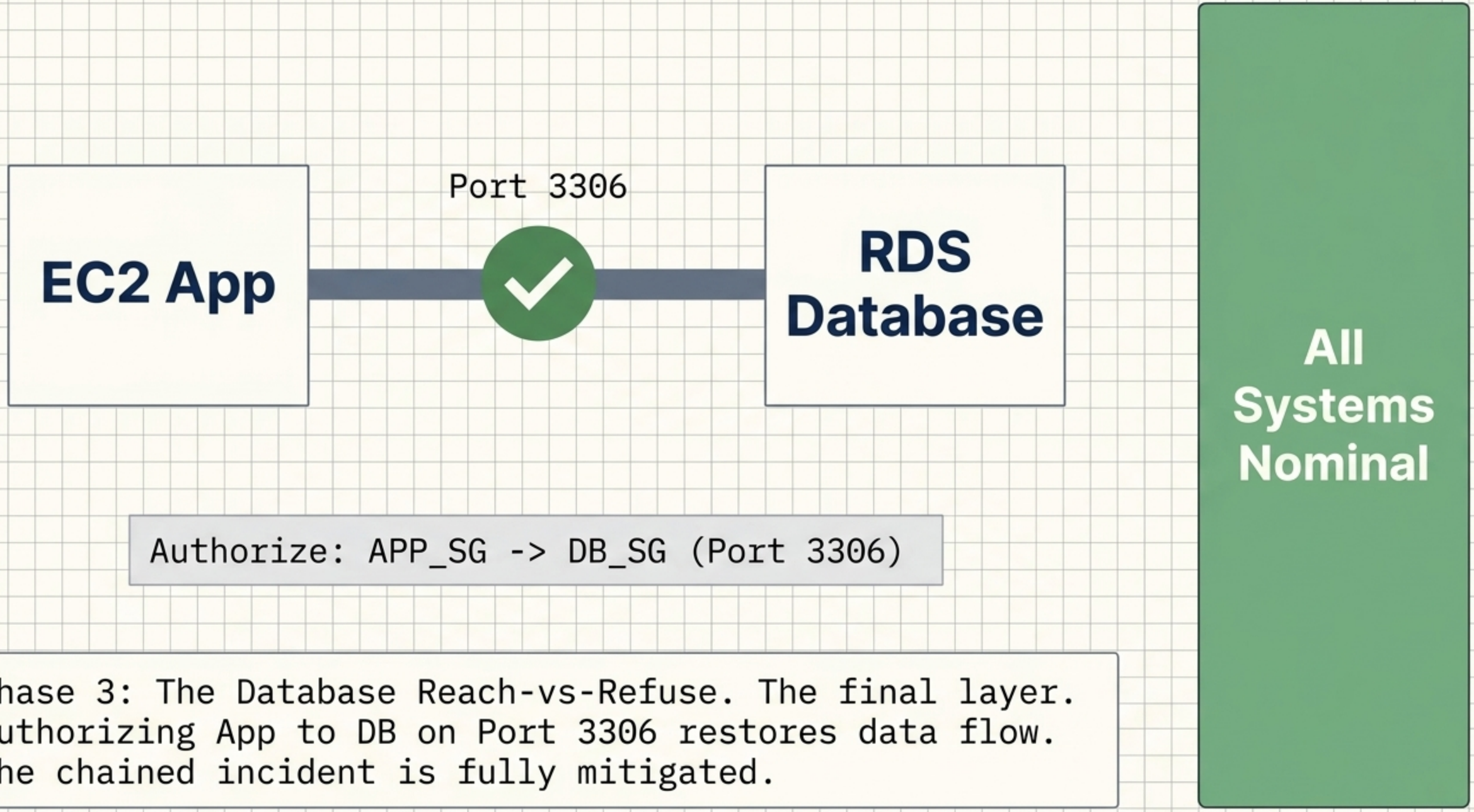
Telemetry

HTTP 200 OK

UI loaded, but DB still cut off (No Data).

Phase 2: Correcting the Application Health Check. A successful network path means nothing if the application reports itself as sick. Fixing the health check path gets the site to 200 OK, but the diagnostic puzzle continues.

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EC2 App
Port 3306
RDS Database
All Systems Nominal
Authorize: APP_SG -> DB_SG (Port 3306)
Phase 3: The Database Reach-vs-Refuse. The final layer. Authorizing App to DB on Port 3306 restores data flow. The chained incident is fully mitigated.

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Writing the Post-Incident Timeline

Accurate documentation of state transitions—and exactly when they occurred—is critical for the post-mortem. A SEV-1 isn't over until the timeline is written.

T+00:00 Incident Paged: 503/504 Outage

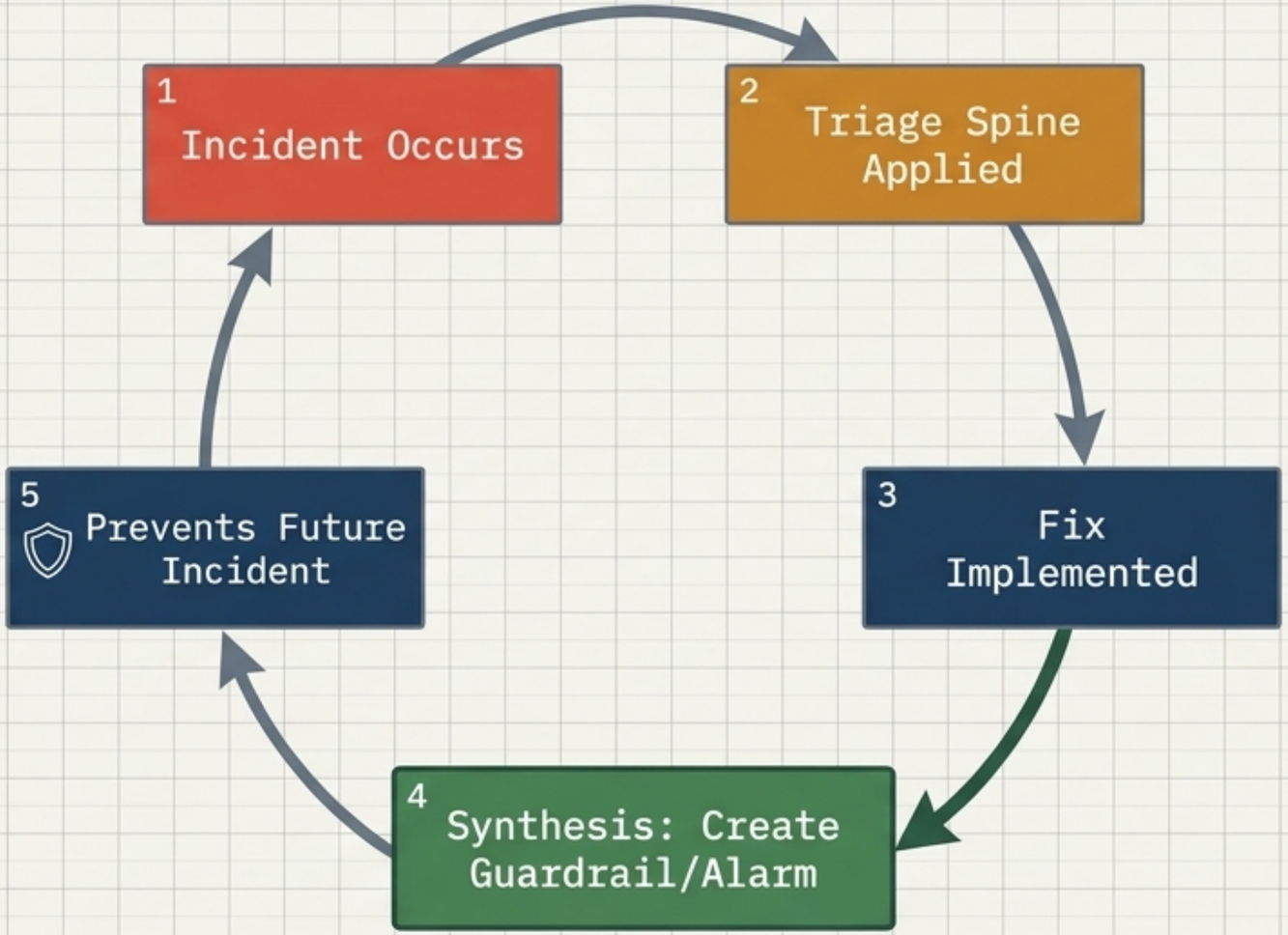
T+15:00 Port 80 authorized; state shifts to Target Unhealthy

T+25:00 Health check path corrected; UI restores (200 OK), DB unreachable

T+35:00 Port 3306 authorized; Full resolution achieved

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The Day 2 Operational Flywheel



Closing the Loop: Every incident must leave behind an alarm, a runbook, or an architectural guardrail. Fixing the issue is only half the jo

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System Core Tenets

> REAL_INCIDENTS_HAVE_LAYERED_CAUSES

Fixing the first anomaly rarely fixes everything. Always verify telemetry after every single change.

> ONE_HYPOTHESIS_AT_A_TIME

Avoid the chaos trap. Isolating variables is the only way to prove a root cause.

> THE_SPINE_SCALES

The fundamental network path walk and DB reach-vs-refuse apply regardless of system complexity.

$ exit 0

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